Polynomial equations

1

a $x = \frac{4}{5}, -3, 2$

b $x = -6, -4, 4$

c $x = -3, 6, \frac{2}{5}$

d $x = 0, \frac{5}{3}$

e $x = 5$

f $x = -10, 5, 0$

g $x = -4, 4, 0$

h $x = 0, \frac{9}{8}, -\frac{9}{8}$

2

a $2x^3\,(x + 20)\,(x - 2) = 0$

b $x = 0, 2, -20$

3

a $\left(4x^2 - 5\right)(x - 3) = 0$

b $x = 3, \frac{\sqrt{5}}{2}, -\frac{\sqrt{5}}{2}$

4

a $(x - 5)\left(x^2 + 6x + 9\right) = 0$

b $x = -3$

Equations reducible to quadratics

5

a

i $p = x^2$

ii $p = 9,\ 25$

iii $x = \pm 3, \pm 5$

b

i $p = x^2$

ii $p = 1,\ 25$

iii $x = \pm 1, \pm 5$

c

i $p = 5x^2$

ii $p = 4,\ 9$

iii $x = \pm\frac{2\sqrt{5}}{5}, \pm\frac{3\sqrt{5}}{5}$

d

i $p = 9x^2$

ii $p = 4,\ 9$

iii $x = \pm 1, \pm\frac{2}{3}$

e

i $p = 2x + 4$

ii $p = -2,\ -8$

iii $x = -3, -6$

f

i $p = 3x + 3$

ii $p = -12,\ -9$

iii $x = -5, -4$

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g

i $p = 4x + 8$

ii $p = -4, -12$

iii $x = -3, -5$

h

i $p = 4x - 4$

ii $p = 7,\ -9$

iii $x = \frac{11}{4}, -\frac{5}{4}$

6

a $x = \pm 1, \pm 2$ **b** $x = \pm 2, \pm 4$ **c** $x = \pm 5, \pm 2\sqrt{6}$ **d** $x = \pm 5, \pm 3$

e $x = -4, -\frac{15}{4}$ **f** $x = -\frac{3}{2}, -\frac{15}{8}$ **g** $x = -\frac{34}{27}, -\frac{5}{3}$ **h** $x = \frac{3}{16}, -\frac{1}{4}$

7

a $x = \pm 3$ **b** $x = \pm 2$ **c** $x = \pm 4$ **d** $x = \pm 5$

e $x = \pm\sqrt{7}$ **f** $x = \pm\sqrt{15}$ **g** $x = \pm\sqrt{14}$ **h** $x = \pm 1$

Applications

8

a Any expression equivalent to $4x^3 - 80x^2 + 300x$.

b $x = 4$

9

a $P = x^3 - 7x^2 - 11x + 9$ **b** $x = 37$

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10

a Answers may vary.
Taking out the central rectangular prism leaves behind a triangular prism of length $23x + 10 - 3x$, height $2x$, and width $2x + 6$.
Putting this together, we have

$$\frac{2x\left(23x + 10 - 3x\right)}{2}\left(2x + 6\right) = 240$$

which we can simplify to

$$x\left(2x + 1\right)\left(x + 3\right) = 12$$

b We can rewrite the equation as

$$2x^3 + 7x^2 + 3x - 12 = 0$$

Using long division or otherwize we can factor this into the form:

$$\left(x - 1\right)\left(2x^2 + 9x + 12\right) = 0$$

The first factor gives the solution $x = 1$, while the second factor has two complex solutions. Since x represents a length (in feet), it must be that $x = 1$.
So the dimensions are as follows:

- The width of the ramp is 8 feet
- The height of the ramp is 2 feet
- The total length of the ramp is 33 feet
- The flat distance between the two sloped sections is 3 feet
- The horizontal length of the shorter sloped section is 12 feet
- The horizontal length of the longer sloped section is 18 feet

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